H5. Frontline Health-Worker Companion for Low-Connectivity Areas

Real-world pain: An ASHA or ANM worker visits villages with a phone, patchy 2G, and responsibility for maternal, child, and chronic-disease follow-ups across hundreds of households. She needs answers from a protocol book she cannot carry, in her own language, oflen with no signal at all.

The problem: Build an assistant that answers frontline health-worker questions from official protocol documents, works **offline or on very poor connectivity**, and knows when to escalate to a doctor.

What we provide:

- A corpus of simplified public-health protocol documents: antenatal care schedule, immunisation schedule, danger signs in pregnancy and newborns, malnutrition screening, hypertension and diabetes follow-up, ORS and fever management
- 200 realistic worker questions in code-mixed language, with hidden expected answers and hidden "must escalate" flags
- A small offline-capable language model and embedding model, pre-cached
- A synthetic household register of 500 families with due-date and immunisation-due information

Assumptions students may make:

- Device is a low-end Android phone; assume tight memory and no GPU
- Connectivity may be zero for hours — the core path must work offline
- The worker is a trained paramedical professional, not a doctor, and not a layperson
- No integration with any government system exists

Desired output:

1. A working question-answering flow **grounded strictly in the provided protocols**, with the protocol section cited in every answer
2. A demonstrated **offline mode**, plus a measured footprint: model size, memory use, and response time on a CPU. This is scored — efficiency is a first-class result here, not an aflerthought
3. A **hard-coded danger-sign escalation layer** that fires before any model reasoning ("if bleeding in third trimester → refer immediately"). `evaluate. py` checks 100% detection on the escalation set — anything less than perfect fails this criterion
4. A daily visit list generated from the household register: who is due for what, prioritised
5. Refusal behaviour on out-of-scope questions ("what dose of this antibiotic?") with a clear redirect to a doctor

   Stretch: voice input in one Indian language, and a sync-when-connected queue for collected data.